

Abhishek Kumar Shakya

Education

2023 – 2027 **B.Tech**, Madan Mohan Malaviya University of Technology, Gorakhpur, India

Major: **Electronics and Communication Engineering**

Minor: **Artificial Intelligence and Machine Learning**

CGPA: **8.29 / 10**

- Relevant coursework: Communication Systems, C Programming, Data Structures and Algorithms, Microprocessors and Applications, Engineering Physics, Digital Electronics and Computer Organization, Digital Signal Processing, Internet of Things(IoT).
- Practical experience with Microcontrollers (Arduino, NodeMCU), Drones, Sensors, Actuators and IoT-based systems.

Experience

Oct 2025 – Jan 2026 **Intern – eSim Project (Remote)**, FOSSEE, IIT Bombay, Mumbai, India

- Working under the guidance of **Mr. Sumanto Kar** (Assistant Project Manager) and **Prof. Prabhu Ramachandran** (Principal Investigator).
- Designing, simulating, and testing analog and digital integrated circuits using the eSim platform.
- **Project Details:** [Click here!](#)

Aug – Sept 2025 **Project Assistant (Remote)**, FOSSEE, IIT Bombay, Mumbai, India

- Worked under the guidance of **Mr. Khushalsingh Rajput** with **Prof. Kannan Moudgalya** as Principal Investigator.
- Assisted in the **FOCAL Project**, contributing to the organization and management of the Animate Hackathon 2025.
- Coordinated event planning, participant management, and communication to ensure smooth execution.

July 2025 – Present **Undergraduate Researcher**, University of Southern California (USC), Los Angeles, USA & Transilvania University of Brasov, Romania

- Conducting research on **Fractional Dynamics Connectome for Brain Age Prediction** for a conference publication.
- Collaborating with **Prof. Paul Bogdan** (USC) and **Dr. Ioana Corina Bogdan** (Transilvania University of Brasov).
- Conducting a literature survey on **Brain Age and the Connectome** and **Brain Microstructural Features**.

May – June 2025 **Summer Intern – Graphic Design**, FOSSEE, IIT Bombay, Mumbai, India

- Created posters, brochures, certificates, social media content, and web interface designs during the Summer Internship program.
- Supported educational and outreach initiatives through effective visual communication.
- **Fellowship details:** <https://fossee.in/fellowship/2025>
- **Certificate:** [View Certificate](#)

Projects

eSim – Digital and Analog IC Designing

Technologies: eSim, Digital and Analog Electronics

Project Link: github.com/abhi26shakya/eSim

- Designed, simulated, and tested analog and digital integrated circuits using the eSim platform.
- Performed circuit analysis and validation through simulation to ensure correct functionality and performance.

Graduate Admission Prediction

Technologies: Python, Pandas, NumPy, Matplotlib, TensorFlow/Keras, Scikit-learn

Project Link: github.com/abhi26shakya/Graduate-Admission-Prediction-using-ANN

- Developed an Artificial Neural Network (ANN) model to predict graduate admission chances based on academic and profile-related parameters.
- Performed data preprocessing, feature scaling, model training, and hyperparameter tuning to improve prediction accuracy.
- Evaluated model performance using regression metrics and analyzed the impact of various factors on admission outcomes.

Customer Churn Rate Prediction

Technologies: Python, Pandas, NumPy, Matplotlib, Scikit-learn

Project Link: github.com/abhi26shakya/Customer-Churn-Rate-Prediction-Project

- Developed a machine learning model to predict customer churn using customer demographic and behavioral data.
- Performed data preprocessing, feature engineering, exploratory data analysis (EDA), and model training to identify factors influencing customer retention.
- Evaluated model performance using classification metrics and generated insights to support data-driven customer retention strategies.

Handwritten Digit Classification

Technologies: Python, NumPy, Pandas, Matplotlib, TensorFlow/Keras

Project Link: github.com/abhi26shakya/Handwritten-Digit-Classification-using-ANN-MNIST-Dataset

- Developed an Artificial Neural Network (ANN) model to classify handwritten digits using the MNIST dataset.
- Implemented data preprocessing, model training, validation, and performance evaluation using deep learning techniques.
- Achieved high classification accuracy and analyzed model performance through visualization and metrics.

Line Follower Bot

Technologies: Arduino Nano, IR Sensor Array, Ultrasonic Sensor, L298N Motor Driver

Project Link: github.com/abhi26shakya/Line-Follower-Bot

- Designed and developed an autonomous line-following robot using an IR sensor array and an ultrasonic sensor for obstacle detection and avoidance.
- Participated in **MazeX** at **Technex'25**, the annual Techfest of **IIT (BHU) Varanasi**.

Drone and IoT Club Website

Technologies: HTML, CSS, JavaScript

Project Link: github.com/Malviyans/Drone-2024

- Developed a responsive website for the Drone and IoT Club using HTML, CSS, and JavaScript.
- Created an interactive, user-friendly interface.

Positions of Responsibility

- 2026 **Event Coordinator Head – Swiftwing's, Drone and IoT Club, MMMUT**
- 2025 **Co-Lead, Drone and IoT Club, MMMUT**
○ Co-led the **Multi-Rotor Drone, Publicity, Media & Outreach**, and **Sponsorship & Merchandise** teams.
- 2023 – Present **Executive Member, Drone and IoT Club, MMMUT**

Extracurricular Activities

- Dec 2024 **Organizing Team Member, Drone and IoT Club, MMMUT – View Certificate**
- Apr – May 2024 **Coordinator, Drone and IoT Club, MMMUT – View Certificate**
- Jan 2024 **Volunteer, Child Rights and You (CRY) – View Certificate**

Technical Skills

- Programming Languages** Java, C, Embedded C, Python, HTML/CSS, JavaScript
- ML/DL Frameworks & Libraries** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, PyTorch, Keras, ANN, CNN, RNN, LSTM, Transformer and Natural Language Processing (NLP)
- Tools & IDEs** GitHub, VS Code, IntelliJ IDEA
- IoT** Internet of Things (IoT), Sensors
- Circuit Simulation** eSim, TinkerCAD, MultiSim
- Graphics Design** Canva, Inkscape, Scribus, Penpot, Synfig Studio

Certifications

- Feb – March 2025 **MazeX at Technex'25, IIT (BHU) Varanasi, View Certificate**
- Jan – Feb 2025 **Ethical Hacking Certificate, Engineer Core in association with IIIT Allahabad, View Certificate**
- Apr – May 2024 **IoT, Drone, 3D Printing, and Artificial Intelligence Workshop, Drone & IoT Club, MMMUT, View Certificate**
- 2022 **International Youth Math Challenge (IYMC), View Certificate**

Interests

AI/ML — Robotics — Neuroscience — Sketching — Football